Nilpotent Sharpness for the Scaled q-Numerical Spectral Constant

Run `fa_banach_001`, lane 0

July 3, 2026

Status

Partial result, likely valid. This packet proves that the constant conjectured in Conjecture 4.2 of O'Loughlin–Rani [1] is globally sharp. It does not prove the conjecture for arbitrary matrices. It proves the conjectured inequality for the 2×2 nilpotent Jordan block and shows that this example already forces the nontrivial branch of the constant.

Source Problem

The source paper is Ryan O'Loughlin and Jyoti Rani, "q-Numerical Ranges and Spectral Sets", arXiv:2603.15536. It defines

$$W_q(A) = \{\langle Ax, y\rangle : \|x\| = \|y\| = 1,\ \langle x, y\rangle = q\}, \qquad \Omega_q(A) = W_q(A)/q,$$

for $0 < |q| \le 1$. Conjecture 4.2 asks whether, for every $n \times n$ matrix A and every polynomial p,

$$\|p(A)\| \le \max\left(1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}}\right) \sup_{z \in \Omega_q(A)} |p(z)|.$$

The source also notes that the term 1 is forced by the constant polynomial.

Conjecture 4.2. *Let A be a $n \times n$ matrix, $0 < |q| \le 1$ and p a polynomial with complex coefficients. Then*

$$\|p(A)\| \le \max\left(1, \frac{2|q|}{1 + \sqrt{1 - |q|^2}}\right) \sup_{z \in \Omega_q} |p(z)|.$$

The term 1 in the maximum arises because a spectral constant is always at least 1, which follows by testing with the constant polynomial $p(z) = 1$.

Proof Intuition

The source paper gives a support-function formula for $\Omega_q(A)$. For a rank-one nilpotent, this formula becomes a one-variable trigonometric optimization. The optimization has a constant value in every direction, so $\Omega_q(J)$ is a disk. Its radius is exactly the reciprocal of the conjectured nontrivial constant:

$$D(q) = \frac{1+\sqrt{1-|q|^2}}{2|q|}, \qquad \frac{1}{D(q)} = \frac{2|q|}{1+\sqrt{1-|q|^2}}.$$

Thus the polynomial $p(z) = z$ gives

$$\|J\| = 1 = \frac{1}{D(q)} \sup_{|z|\le D(q)} |z|$$

whenever this reciprocal is the larger branch of the conjectured maximum. The branch switch occurs at $D(q) = 1$, equivalently $|q| = 4/5$.

Main Result

Theorem 1. *Let*

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad on\ \mathbb{C}^2, \qquad 0 < |q| \le 1,$$

and put $r = |q|$, $s = \sqrt{1-r^2}$. Then

$$\Omega_q(J) = \{z \in \mathbb{C} : |z| \le D_r\}, \qquad D_r = \frac{1+s}{2r}.$$

Consequently the conjectured inequality of O'Loughlin–Rani holds for J. Moreover, the conjectured universal constant is sharp: the lower branch 1 is forced by constant polynomials, and for $r \ge 4/5$ the polynomial $p(z) = z$ forces the branch

$$\frac{2r}{1+\sqrt{1-r^2}}.$$

Proof. The source paper observes that $\Omega_q = \Omega_{|q|}$, so we may take $q = r > 0$. Its support-function formula specializes to

$$h_{\Omega_r(A)}(e^{i\theta}) = \sup_{\|x\|=1} \left(\Re\langle e^{-i\theta}Ax, x\rangle + \frac{s}{r}\|P_{x^\perp}(e^{-i\theta}Ax)\| \right),$$

where $s = \sqrt{1-r^2}$.

Now take $A = J$. Write a unit vector as

$$x = (\cos t, e^{i\phi}\sin t), \qquad 0 \le t \le \pi/2.$$

The phase ϕ can be chosen, for each θ, to make $\Re\langle e^{-i\theta}Jx, x\rangle = |\langle Jx, x\rangle|$. Also

$$|\langle Jx, x\rangle| = \sin t\cos t, \qquad \|P_{x^\perp}Jx\| = \sqrt{\|Jx\|^2 - |\langle Jx, x\rangle|^2} = \sin^2 t.$$

Therefore the support function is independent of θ and equals

$$\sup_{0\le t\le\pi/2} \left(\sin t\cos t + \frac{s}{r}\sin^2 t \right).$$

Using $\sin t\cos t = \frac{1}{2}\sin 2t$ and $\sin^2 t = \frac{1}{2}(1-\cos 2t)$, this supremum is

$$\frac{s}{2r} + \frac{1}{2}\sup_u \left(\sin u - \frac{s}{r}\cos u \right) = \frac{s}{2r} + \frac{1}{2}\sqrt{1+\frac{s^2}{r^2}} = \frac{s}{2r} + \frac{1}{2r} = D_r.$$

Hence the support function is the constant D_r, which proves that $\Omega_r(J)$, and therefore $\Omega_q(J)$, is the closed disk of radius D_r.

It remains to check the functional-calculus inequality for this model. Since $J^2 = 0$, every polynomial satisfies

$$p(J) = aI + bJ, \qquad a = p(0), \quad b = p'(0).$$

Thus

$$\|p(J)\| \le |a| + |b|.$$

On the disk $|z| \le D_r$,

$$\sup_{|z| \le D_r} |a + bz| = |a| + D_r|b|.$$

If $D_r \ge 1$, equivalently $r \le 4/5$, the conjectured constant is 1, and

$$\|p(J)\| \le |a| + |b| \le |a| + D_r|b| = \sup_{z \in \Omega_q(J)} |p(z)|.$$

If $D_r \le 1$, equivalently $r \ge 4/5$, the conjectured constant is $1/D_r$, and

$$\|p(J)\| \le |a| + |b| \le \frac{1}{D_r}(|a| + D_r|b|) = \frac{1}{D_r} \sup_{z \in \Omega_q(J)} |p(z)|.$$

This proves the conjectured inequality for J.

Finally, constant polynomials force every spectral constant to be at least 1, as already noted in the source paper. For $p(z) = z$, one has $\|p(J)\| = \|J\| = 1$ and

$$\sup_{z \in \Omega_q(J)} |p(z)| = D_r.$$

Thus any universal constant valid for all matrices must be at least $1/D_r$. For $r \ge 4/5$, $D_r \le 1$, so $1/D_r$ is exactly the nontrivial branch

$$\frac{2r}{1 + \sqrt{1 - r^2}}.$$

The conjectured constant is therefore sharp. □

Verification Notes

The scalar threshold is

$$D_r = 1 \quad \Longleftrightarrow \quad 1 + \sqrt{1 - r^2} = 2r \quad \Longleftrightarrow \quad r = \frac{4}{5}.$$

The included script `code/verify_nilpotent_radius.py` checks the identity $C(r)D_r \ge 1$, with equality on the nontrivial branch $r \ge 4/5$, for a dense grid of r-values.

Novelty and Literature Check

The lightweight run indexes were searched for arXiv:2603.15536, the exact title, q-numerical range, scaled q-numerical range, spectral set, and Crouzeix keywords. No existing packet for this source paper was found. External web search on July 3, 2026 used the exact title and phrases "scaled q-numerical range Crouzeix conjecture", "q-numerical range spectral set $2|q|$", and "$W_q(A)/q$ Crouzeix". This found the source paper but no later explicit solution or sharpness note for Conjecture 4.2. This is a bounded novelty check, not a publication-level literature review.

Limitations

This packet does not settle Conjecture 4.2 for general matrices. It identifies a sharp extremal model and proves the conjectured inequality in that model. The main value is that it proves the constant in the conjecture cannot be lowered.

Human-Review Recommendation

Review as a likely valid partial result and sharpness lemma. The key step to check is the support-function calculation for the 2×2 nilpotent Jordan block; it is a direct specialization of the source paper's formula.

References

[1] Ryan O'Loughlin and Jyoti Rani, *q-Numerical Ranges and Spectral Sets*, arXiv:2603.15536, 2026. `https://arxiv.org/abs/2603.15536`.